

Poisoning the Threshold-Calibration Stage in Federated IoT Anomaly Detection: A Policy-Differentiated Vulnerability Analysis

Salaheddine Naslouby^{1,*} , Mohamed Lachgar², and Hamid Hrimech¹

¹ Hassan First University, National School of Applied Sciences (ENSA), LAMSAD Laboratory, Berrechid, Morocco

*naslouby.salahe@gmail.com

hamid.hrimech@uhp.ac.ma

² Cadi Ayyad University, Higher Normal School (ENS), L2IS Laboratory, Marrakech, Morocco

m.lachgar@uca.ac.ma

Abstract. Federated learning (FL) defenses for IoT anomaly detection largely focus on training and aggregation, while the post-training *threshold-calibration stage* lets each client autonomously collect benign calibration scores and derive its detection threshold. This paper characterizes that stage under a score-level contamination model: one compromised client replaces a fraction f of its calibration buffer with tail-sampled benign reconstruction scores, shifting the threshold without altering model weights, training data, gradients, aggregation, test scores, or labels. Across Global, Local, and Cluster threshold policies on the N-BaIoT nine-device federation, the raising attack achieves a 100% pass rate on the pre-specified materiality gate at all non-zero fractions and degrades victim true-positive rates by 2.4–7.9 percentage points (4.9–7.9 pp at $f=0.4$). The lowering attack increases false-alarm burden for Local and Cluster, raising cross-client CV(FPR) by 0.05–0.06 relative to the clean baseline at $f=0.4$ under fixed-budget replacement; the associated rise in victim TPR reflects threshold displacement, not a security benefit. Global bounds the lowering attack through $1/N$ dilution. Policy choice governs both magnitude and propagation: Local confines threshold harm to the compromised client, Cluster can transmit small co-cluster perturbations, and Global dilutes but propagates shifts federation-wide. The results are bounded to N-BaIoT and score substitution; they characterize a calibration-channel vulnerability mechanism, not a complete traffic-generation attack.

Keywords: Federated Learning · Anomaly Detection · IoT Security · Poisoning Attack · Threshold Calibration · Vulnerability Analysis

1 Introduction

Federated learning (FL) is commonly used for IoT anomaly detection: devices contribute to a shared model without centralizing raw traffic, and personalized

detection thresholds let each node adapt to its own traffic distribution [15]. Security work on federated anomaly detection has concentrated on the *training phase*: Byzantine clients injecting adversarial gradients, label-flipping poisoning, or model-inversion attacks [4,5,19]. Aggregation-phase defenses such as FLTrust, Krum, and Trimmed-Mean target exactly this surface [6,7,20]. Recent FL-IDS work adds personalized, non-IID-robust, knowledge-distillation, and anti-poisoning designs [3,9,14,18], all targeting the training or aggregation surface rather than the post-training calibration buffer.

A post-training coverage gap. The *threshold-calibration stage* is the post-training, pre-deployment step in which each client collects a window of benign traffic, computes local reconstruction scores, and derives its detection threshold at a fixed quantile q . Training- and aggregation-phase defenses complete before calibration begins and cannot observe the calibration buffer. If a single client’s benign calibration buffer can be contaminated with benign scores drawn from the upper (or lower) tail of that client’s own distribution, the resulting threshold shifts without any change to model weights, training data, labels, or gradients. No claim is made that a dedicated calibration-stage defense is impossible, only that the training- and aggregation-phase defenses studied in the FL-IDS literature do not cover this stage.

Study focus. This paper identifies threshold calibration as an underexplored attack surface and characterizes its vulnerability under a score-level contamination model. Concretely, an adversary with access to a single client’s calibration buffer replaces a fraction f of entries with tail-sampled scores drawn from the same client’s benign distribution (*calibration-stage poisoning*). The evaluation covers three threshold-policy families—Global (federation-wide mean), Local (per-client), and Cluster (cluster-averaged)—across all nine physical-device clients in the N-BaIoT dataset [13] at four injection fractions.

Calibration-stage poisoning is not simply data poisoning relocated to calibration data (Table 1). It occurs strictly after model convergence, so training-phase defenses have already completed; it exploits a trust model specific to that phase, since calibration data is collected autonomously with no central validation gate; and its effect depends on the threshold policy, so the same injection fraction produces different shifts and different propagation under Global, Local, and Cluster.

Contributions.

1. The analysis isolates the post-training threshold-calibration stage as a distinct attack surface: contaminating one client’s benign calibration scores can shift thresholds while model weights, training data, aggregation, test scores, and labels remain fixed (§3).
2. The evaluation presents a controlled vulnerability characterization under score-level calibration contamination across three threshold policies, showing that the raising attack degrades victim detection rates by 2.4–7.9 pp across non-zero injection fractions (4.9–7.9 pp at $f=0.4$) with 95% bootstrap confidence intervals excluding zero (§5).
3. The results document policy-differentiated behavior: Local gives the largest per-victim threshold shifts with no cross-client threshold propagation; Clus-

ter allows small intra-cluster threshold perturbations; Global dilutes shifts but propagates them federation-wide; lowering is secondary and conditionally bounded, especially under Global (§5).

Scope. All claims are bounded to the nine-device N-BaIoT setting with autoencoder detectors and FedAvg aggregation, and the poisoning mechanism operates at the score level; no claim is made about deployability, evasion guarantees, or generalization beyond this setting. Defense design against calibration-stage poisoning is out of scope, though §6 discusses properties a defense would need to satisfy.

2 Background and Related Work

Federated anomaly detection for IoT. Federated intrusion and anomaly detection for IoT has moved beyond plain FedAvg training [3,9,14,15,18]. These studies improve training-time collaboration and robustness metrics, while thresholded autoencoder pipelines still rely on a benign post-training calibration step. The N-BaIoT benchmark [13] provides labeled benign and Mirai/BASHLITE traffic for nine physical devices, establishing a widely used evaluation platform for FL-IDS research.

Defenses. Existing defenses target the *training phase*. Byzantine-robust rules, including Krum [6], Trimmed-Mean, and Median [20], filter or downweight outlier gradients before aggregation. FLTrust [7] anchors aggregation to a trusted server dataset. These defenses operate on gradients or model updates; they complete before the threshold-calibration stage begins. Recent FL-IDS defenses extend this training-side view with personalized poisoning defenses, knowledge-distillation verifiers, and probe-guided anti-poisoning pipelines under non-IID IoT data [9,14,18]; the defenses reviewed here do not observe the later client-local calibration buffer.

Threshold calibration and poisoning. By construction, a detection threshold at fixed quantile q is sensitive to the scores admitted to the calibration distribution: tail-high contamination raises the boundary (missed detections); tail-low contamination lowers it (false alarms). Kloft & Laskov [10] study online poisoning in centralized anomaly detection via a continuous training stream, assuming a trusted aggregator or persistent model-update access—both absent from FL’s post-training calibration stage, where calibration data never leaves the client and the pipelines considered here perform no server-side sanitization. FL model poisoning [2] also targets the gradient/weight surface. No prior work in the reviewed literature isolates post-training threshold calibration as the attacked channel in federated threshold-based IoT anomaly detection under otherwise-unchanged training, model, aggregation, and test artifacts (Table 1). This is not a new poisoning taxonomy; the contribution is isolating this channel and showing policy-dependent victim effects and threshold propagation under a controlled score-level intervention. The closest related work treats the calibration/thresholding pipeline as trusted: Laridi et al. [11] propose summary-statistics-based federated thresholding without adversarial contamination, and

Table 1. Positioning relative to adjacent poisoning literature.

Prior work type	Attack surface	What changes	Distinction from this work
Training-data poisoning [5,19]	Training set / labels	Model parameters	Calibration-stage poisoning leaves training data and model unchanged
FL model poisoning [2,4]	Client gradients	Model weights	Needs no gradient access; acts after convergence
Byzantine aggregation [6,20]	Aggregation inputs	Server aggregate	Does not touch aggregation; harm is post-aggregation
Centralized AD poisoning [10]	Central train stream	Decision boundary	No server-side sanitizer in the FL calibration pipelines considered
Threshold / model personalization [8,11,17]	Threshold policy	Per-client/cluster τ	Studies utility, not adversarial contamination
This work	Calibration scores	Threshold τ only	Score-level, post-training, single-client

Sáez-de-Cámara et al. [16] propose a clustered federated IoT anomaly-detection architecture assuming a benign calibration pipeline; calibration-stage poisoning instead treats client-local benign calibration as the attacked channel.

Threshold policies. The Global/Local/Cluster threshold policies evaluated here follow the client-partitioning and personalization structures used in federated personalization work [8,11,17] (defined precisely in §4); their security implications have not been studied under calibration-stage adversaries.

3 Threat Model

Attack surface. The threshold-calibration stage occurs after global model convergence and before threshold-dependent detection begins (Fig. 1). Each client i collects a benign calibration set $\mathcal{C}_i = \{s_1, \dots, s_{n_i}\}$ of reconstruction scores from its own traffic, then sets its local threshold $\tau_i = \text{quantile}(\mathcal{C}_i, q)$ at a fixed quantile q (here $q=0.95$). Depending on the federation’s threshold policy, τ_i may be used directly (Local), averaged across clients (Global), or averaged within a calibration-score cluster (Cluster).

The training-phase and aggregation-phase defenses considered in this study do not monitor or validate the calibration buffer: it is collected autonomously by the client after the aggregation round completes.

Adversary model. The adversary controls a single compromised client v . Under Local policy, only client v ’s own detection degrades; under Global policy,

all federation members receive the contaminated threshold; under Cluster, co-cluster members may receive a perturbed threshold. The term “victim” refers to client v itself when measuring per-client detection harm, and “non-victim clients” refers to uninvolved federation members.

The adversary’s access is restricted to the score level and to a single client: it can read and modify its own local calibration buffer before threshold computation. It cannot modify training data, training labels, test scores, test labels, model weights, gradients, the aggregation procedure, other clients’ data, client eligibility, or the threshold-policy definition (Global/Local/Cluster are fixed by the federation). The reservoir is drawn exclusively from the same client’s own benign calibration split (§4): attack-labeled samples, test scores, and training scores never enter it. This capability is weaker than that of model-poisoning adversaries, which require gradient injection. The adversary operates at the score level; generating traffic that achieves a target reconstruction score is out of scope (end-to-end realizability is discussed in §6).

Attack mechanics. The evaluation considers two attack objectives:

Raise. Threshold-Raise (primary): $f=0$ is the unpoisoned clean baseline ($m=0$, no replacement); for $f>0$, replace $m = \max(1, \text{round}(fn_v))$ positions in $\mathcal{C}_v$ with scores drawn with replacement from the top-10% tail of $\mathcal{C}_v$ (High-Score-Benign source; this fixed-budget replacement rule is Replace-Fixed-Budget). All replacement scores are benign-origin reconstruction scores drawn from the victim’s own clean calibration distribution; AUROC is invariant by construction (test scores and labels are unchanged under poisoning). A raised threshold τ_v increases the detection threshold, suppressing alarms at the victim site.

Lower. Threshold-Lower (secondary, conditional): replace m positions with scores from the bottom-10% tail (Low-Score-Benign). A lowered threshold increases the false-positive rate (alarm burden) and FPR disparity; the resulting $\Delta\text{TPR}>0$ reflects threshold displacement and is a secondary indicator of operating-point shift.

Negative control. Random-Benign: replacement scores are sampled uniformly from the full calibration distribution rather than a tail. Any effect attributable to score contamination rather than position perturbation should vanish in this condition.

Scope boundary. The analysis characterizes the vulnerability at the score level. A complete end-to-end attack would additionally require generating network traffic whose autoencoder reconstruction score lands in the target tail—this is future work (§6). Claims are bounded to the single-client, single-compromise scenario; multi-client collusion is not evaluated.

4 Experimental Protocol

4.1 Dataset and Federation

The evaluation uses N-BaIoT [13], a labeled IoT network traffic dataset covering nine physical devices spanning five device types (doorbell, baby monitor, secu-

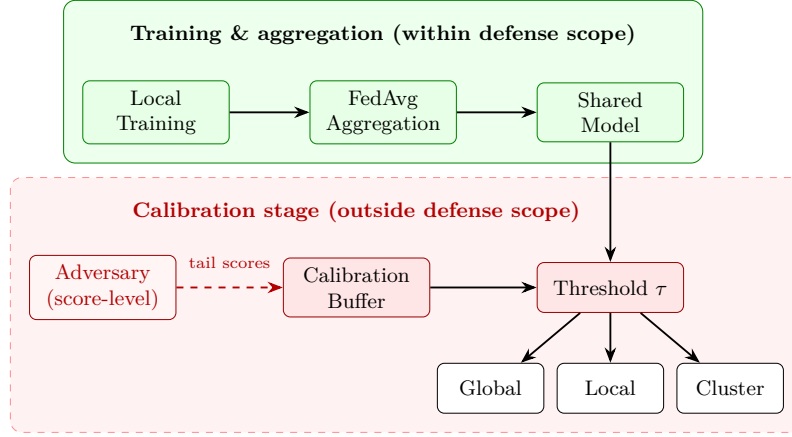


Fig. 1. Attack surface: threshold calibration occurs after FL training. Green solid boxes mark training/aggregation phases; the dashed red region marks calibration, outside the observed defense scope. Score-level access to one client’s calibration buffer can shift τ without modifying model weights, gradients, or labels.

rity camera, thermostat, webcam). Each device constitutes one federation client. All nine satisfy the ≥ 100 -sample calibration eligibility threshold (size range in Table 2; per-device sizes in the released split manifest). Each device’s benign stream is partitioned chronologically into a 60% train / 1% gap / 20% calibration / 1% gap / 18% test split; the two gaps avoid temporal leakage. Training and calibration use benign traffic only, and the attack never touches test data, test labels, or training labels. Attack-labeled records are scaled with each device’s training-fitted scaler and used only as the positive class for TPR; they never enter training, calibration, poisoning reservoirs, or injected buffers. The federation trains a shared autoencoder on benign traffic using FedAvg [12]; each client then scores its own traffic with the aggregated model and enters the threshold-calibration stage at a fixed quantile $q=0.95$. Table 2 lists the model and training configuration; injection fractions, seeds, and the reservoir are detailed below.

4.2 Threshold Policies

Three threshold-policy families are evaluated, covering the main design points in the personalization literature:

Global.

$$\tau_i^{\text{local}} = \text{quantile}(\mathcal{C}_i, q), \quad \tau^{\text{global}} = \frac{1}{N} \sum_{i=1}^N \tau_i^{\text{local}}.$$

Global assigns the same τ^{global} to every client; a single poisoned local threshold contributes only $1/N$ of the shared threshold shift.

Local. Each client retains $\tau_i = \tau_i^{\text{local}}$. Shifts are confined to the compromised client; no cross-client threshold propagation.

Table 2. Experimental configuration: autoencoder architecture, federated training hyperparameters, and calibration settings.

Component	Configuration
<i>Autoencoder architecture</i>	
Input features	115 (N-BaIoT statistical traffic features)
Encoder layers	115 $\rightarrow$ 80 $\rightarrow$ 40 $\rightarrow$ 20; decoder mirrors
Activation	ReLU (no batch normalization)
Loss	Mean squared reconstruction error
Optimizer	Adam, learning rate 0.001, weight decay 0
<i>Federated training (FedAvg)</i>	
Local epochs	1
Aggregation	Sample-count-weighted mean
Rounds	40 initial, 150 max (conv.: $\Delta\text{loss} \leq 0.005$ over 10)
Batch size	256
<i>Calibration (per-device sizes in released split manifest)</i>	
Benign samples / device	2,622 (Ecobee) – 35,048 (Philips)
Threshold quantile q	0.95

Cluster. Clients are grouped on a standard-scaled 4-feature fingerprint (mean, std, skew, p_{95} of $\mathcal{C}_i$) via k -means++ [1] ($K=3$, random seed 42; clustering recomputed after each attack draw so a poisoned fingerprint can change the compromised client’s cluster assignment). Within each cluster, the shared threshold is the cluster mean; the higher seed-to-seed variance in Cluster CV(FPR) is consistent with this cluster-assignment sensitivity. A compromised client may perturb the threshold for co-cluster members (*intra-cluster threshold perturbation*); clients in other clusters are unaffected. Tail injection raises the mean, p_{95} , and dispersion components of the fingerprint, typically displacing the victim into an effectively singleton cluster whose mean collapses to the victim’s own poisoned local threshold, producing near-Local victim-level shifts while co-cluster members see near-zero perturbation.

4.3 Attack Parameterization

For each (policy, attack objective, injection fraction $f \in \{0.0, 0.1, 0.2, 0.4\}$) combination, the protocol runs 10 paired seeds: each seed fixes a training configuration and an independent poisoning draw. A clean copy of the calibration array is preserved; only the poisoned copy is used for threshold computation. The four source-objective pairs are: (1) High-Score-Benign + Threshold-Raise (primary raising attack), (2) Low-Score-Benign + Threshold-Lower (lowering attack), (3) Random-Benign + Threshold-Raise (objective-matched negative control for the raising attack), and (4) Random-Benign + Threshold-Lower (negative control for the lowering attack). The attack is applied to each of the nine clients as victim in turn (single-client scope), yielding 4 source-objective pairs $\times$ 3 threshold policies $\times$ 4 injection fractions $\times$ 9 victim clients $\times$ 10 seeds = 4,320 experimental cells total. Training seed $i \in \{0, \dots, 9\}$ is paired with poisoning

seed $i+100$ and analysis seed $i+300$, separating training, calibration sampling, and bootstrap resampling stochasticity.

4.4 Statistical Analysis

Gate 1 (material shift, primary gate). Threshold shift $\Delta\tau_v = \tau_v^{\text{poisoned}} - \tau_v^{\text{clean}}$ is computed per seed. A (victim, seed) cell is *material* when $|\Delta\tau_v| > \delta\tau_v$ and the sign is consistent with the attack objective, where the materiality threshold is $\delta\tau_v = \max(0.1 \times \text{IQR}(\mathcal{C}_v^{\text{clean}}), 0.01 \times \text{median}_j(\text{IQR}(\mathcal{C}_j^{\text{clean}})))$. A *seed passes* when $\geq 5/9$ of the nine clients, each taken as victim in turn, are material. A (policy, fraction) configuration *passes Gate 1* when $\geq 8/10$ seeds pass.

Gate 2 (directional excess over control). Bootstrap 95% CI on mean $\Delta\tau$ excludes zero for the primary strategy but spans zero for the negative control (Random-Benign).

Gate 3 (downstream harm). Mean victim TPR change, ΔTPR , is directionally consistent (negative for raising, positive for lowering) across $\geq 8/10$ seeds.

Confidence intervals. All CIs are 95% percentile bootstrap over 10 seed-level aggregates, with $B=10,000$ resamples. Primary claim gates are evaluated over these paired seed-level aggregates; the per-victim-seed cells are descriptive support, not independent units. With ten paired seeds, percentile-bootstrap intervals are approximate uncertainty estimates rather than exact coverage guarantees, and may deviate from nominal at the tails. A two-sided Wilcoxon signed-rank test on ten seed-level mean victim ΔTPR values (raising, $f=0.4$) gives $W=0$, $p=0.002$ (exact) for all three policies.

Gate constant justification. The 10% IQR coefficient for Gate-1 materiality ($0.1 \times \text{IQR}$) exceeds the calibration instability observed in the clean baseline while remaining below the shift implied by replacing 10% of a calibration buffer with top-decile scores; the 1% cross-client floor ($0.01 \times \text{median}_j(\text{IQR})$) prevents degenerate clients with near-zero score spread from counting as material on arbitrarily small shifts. Both constants were locked from clean-artifact statistics before any poisoned run was executed.

4.5 Isolation Sanity Check

To confirm that the attack is confined to the calibration stage, the protocol verifies that (i) AUROC is invariant under poisoning for all cells, (ii) per-client rank ordering of clean anomaly scores is unchanged, and (iii) model weights before and after calibration are bitwise identical. All three hold in all 4,320 cells. These quantities are invariant by construction, so the check establishes correct isolation of the intervention rather than providing independent evidence.

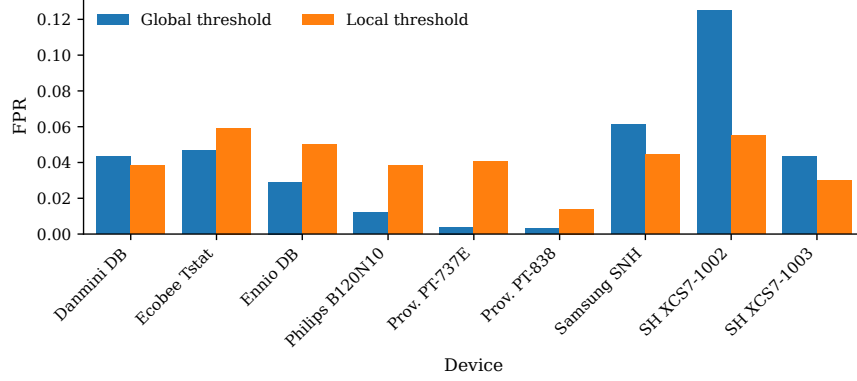


Fig. 2. Clean-baseline FPR heterogeneity is larger under the shared Global threshold than under Local calibration (seed 0). All-seed dispersion, including Cluster, is reported in Table 3. Device labels: DB, doorbell; Tstat, thermostat; SH, SimpleHome; Prov., Provision.

5 Results

5.1 Clean-Baseline Policy Profile

Table 3 reports clean-baseline dispersion across all ten seeds. Lower CV(FPR) means more uniform per-client false-positive treatment: Local is lowest (0.333 ± 0.034), Global highest (0.905 ± 0.039), and Cluster intermediate (0.615 ± 0.143) with cluster-assignment sensitivity. Pairwise CV(FPR) gaps are positive with bootstrap CIs excluding zero: G–L 0.572 [0.537, 0.607], G–C 0.291 [0.214, 0.358], and C–L 0.281 [0.201, 0.374] (unrounded per-seed aggregates); Fig. 2 shows the underlying per-client FPR distributions. Lower dispersion does not imply uniformly better detection: Global retains the best P10 Macro-F1—the 10th percentile of per-client macro-F1, a worst-case detection-quality metric—at 0.398 against Local’s 0.335. Table 3 therefore reports dispersion, detection quality, and mean FPR/TPR rather than a single summary.

Table 3. Clean baseline policy profile on N-BaIoT ($n=10$ seeds, 9 clients). CV: coefficient of variation; BA: balanced accuracy; P10 F1: P10 Macro-F1; FPR/TPR: client/seed means; bold: best; * std ≈ 0.0002 .

Policy	CV(FPR)	CV(TPR)	Worst BA	P10 F1	FPR	TPR
Global	0.905 ± 0.039	0.255 ± 0.030	0.657 ± 0.008	0.398 ± 0.053	0.041	0.793
Local	0.333 ± 0.034	0.296 ± 0.036	0.655 ± 0.005	0.335 ± 0.058	0.043	0.777
Cluster	0.615 ± 0.143	0.313 ± 0.009	0.656 ± 0.004	$0.299 \pm 0.000^*$	0.043	0.742

5.2 Raising Attack: Primary Results

Table 4 and Fig. 3(a) report threshold shift and downstream detection harm for High-Score-Benign + Threshold-Raise.

Table 4. Calibration-stage poisoning shifts thresholds under all policies (N-BaIoT, $n=10$ seeds). $\Delta\tau$: mean shift [95% CI]; G1: Gate-1 seed pass rate; ΔTPR is victim detection-rate change. Lowering rows ($\dagger$) use $\Delta\text{CV}(\text{FPR})$ as the primary harm metric; Neg. Control: Random-Benign at $f=0.4$; bold=zero G1.

Policy	Attack	f	$\Delta\tau$ [95% CI]	G1	ΔTPR (pp)	$\Delta\text{CV}(\text{FPR})$
Global	Raise	0.1	+0.033 [+0.031, +0.035]	100%	−2.4	—
		0.2	+0.062 [+0.059, +0.065]	100%	−4.6	—
		0.4	+0.100 [+0.096, +0.105]	100%	−6.1	—
Local	Raise	0.1	+0.298 [+0.281, +0.314]	100%	−6.0	—
		0.2	+0.558 [+0.535, +0.581]	100%	−7.3	—
		0.4	+0.902 [+0.864, +0.944]	100%	−7.9	—
Cluster	Raise	0.1	+0.253 [+0.231, +0.277]	100%	−3.9	—
		0.2	+0.504 [+0.475, +0.532]	100%	−4.9	—
		0.4	+0.899 [+0.859, +0.941]	100%	−4.9	—
Global	Lower $\dagger$	0.1	−0.006 [−0.006, −0.006]	60%	+0.8	−0.007
		0.2	−0.010 [−0.011, −0.010]	100%	+1.1	−0.011
		0.4	−0.018 [−0.020, −0.017]	100%	+2.1	−0.019
Local	Lower $\dagger$	0.1	−0.053 [−0.057, −0.049]	100%	+2.8	−0.002
		0.2	−0.094 [−0.100, −0.088]	100%	+5.9	+0.005
		0.4	−0.164 [−0.176, −0.154]	100%	+11.8	+0.053
Cluster	Lower $\dagger$	0.1	−0.047 [−0.065, −0.031]	100%	+2.2	+0.007
		0.2	−0.100 [−0.115, −0.084]	100%	+5.4	+0.011
		0.4	−0.154 [−0.167, −0.142]	100%	+10.3	+0.060
Global	Neg. Ctrl	0.4	−0.000 [−0.001, +0.001]	0%	+0.3	—
Local	Neg. Ctrl	0.4	−0.002 [−0.009, +0.006]	0%	+0.9	—
Cluster	Neg. Ctrl	0.4	−0.009 [−0.022, +0.003]	0%	+0.4	—

All three policies clear every gate. The seed-level Gate-1 pass rate is 100% (10/10 seeds) at every non-zero fraction for all three policies. For Local and Cluster, all nine clients are individually material in every seed. For Global at $f=0.1$, two devices (Provision PT-737E and PT-838) are non-material in 2/10 and 4/10 seeds respectively, but the victim-majority condition ($\geq 5/9$ per seed) still holds because the remaining seven clients are consistently material. Bootstrap CIs on $\Delta\tau$ exclude zero and do not overlap the Random-Benign control. For that control, CIs span zero, the Gate-1 pass rate is 0%, and $|\Delta\text{TPR}| \leq 1$ pp at every fraction and policy (Table 4 reports representative $f=0.4$ rows; the artifact contains the full near-null matrix). The targeted effects are therefore attributable to tail contamination, not position perturbation alone.

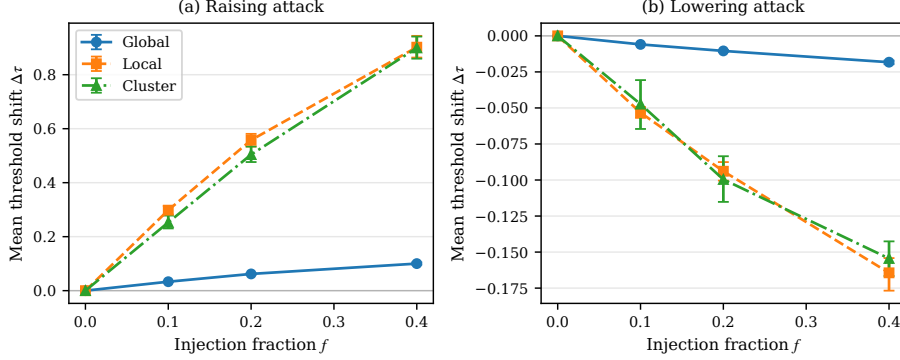


Fig. 3. Mean threshold shift $\Delta\tau$ vs. injection fraction f for raising (a) and lowering (b) attacks. Error bars are 95% bootstrap CIs over 10 seeds; Global is diluted by $1/N$, and Random-Benign controls are near-null (Table 4). Panel (b) uses the legend of panel (a).

Policy magnitude. Raising-attack magnitude differs by policy. Local and Cluster exhibit the largest threshold shifts: at $f=0.4$, $\Delta\tau = +0.902$ [$+0.864, +0.944$] and $+0.899$ [$+0.859, +0.941$] respectively. Global shows a diluted but still material shift: $+0.100$ [$+0.096, +0.105$] at $f=0.4$. The dilution is expected, since Global’s threshold is the federation mean and a single-client shift is divided by $N=9$, but the residual magnitude ($\approx 10\%$ of the local shift) still produces measurable downstream harm.

Downstream harm (Gate 3). Mean victim ΔTPR is negative and directionally consistent across all seeds for all three policies. At $f=0.4$: Local -7.9 pp, Cluster -4.9 pp (unchanged from $f=0.2$), Global -6.1 pp. Downstream harm is not proportional to $\Delta\tau$: because each policy’s clean threshold sits at a different point on the victim’s score distribution, the same displacement traverses different amounts of probability mass, so Global’s smaller shift can still produce a larger TPR loss. Absolute clean $\rightarrow$ poisoned victim TPR at $f=0.4$ (Table 3 clean column): Global $0.793 \rightarrow 0.732$, Local $0.777 \rightarrow 0.698$, Cluster $0.742 \rightarrow 0.693$. Under Global, all eight non-victims receive the identical $\Delta\tau$ (Table 5), but their TPR was not recomputed, so non-victim impact is inferred, not measured. Local/Cluster non-victim threshold propagation is bounded by construction.

5.3 Lowering Attack: Secondary Results (Conditional)

Fig. 3(b) shows threshold shift for Low-Score-Benign + Threshold-Lower. The adversary’s goal is to increase false-alarm burden; the $\Delta\text{TPR} > 0$ below is a threshold-displacement indicator, not a security benefit.

Local and Cluster succeed; Global is marginal. Local achieves $\Delta\tau = -0.164$ [$-0.176, -0.154$] at $f=0.4$ with elevated FPR disparity ($\Delta\text{CV}(\text{FPR}) = +0.053$); Gate-3 $\Delta\text{TPR} = +11.8$ pp is consistent with threshold displacement. Cluster similarly achieves -0.154 [$-0.167, -0.142$] at $f=0.4$ with $\Delta\text{CV}(\text{FPR}) =$

+0.060 and $\Delta\text{TPR} = +10.3$ pp. Global is structurally bounded: at $f=0.4$, $\Delta\tau = -0.018$ $[-0.020, -0.017]$, Gate 1 passes only at $f \geq 0.2$, and $\Delta\text{TPR} = +2.1$ pp.

FPR disparity (operational harm). The operational harm metric is $\Delta\text{CV}(\text{FPR})$ (dimensionless): +0.053 (Local, 10/10 seeds) and +0.060 (Cluster, 8/10 seeds) at $f=0.4$, while Global shows -0.019 because under a shared threshold the FPR rises proportionally across clients, which reduces relative disparity. Under a shared threshold, $\Delta\text{CV}(\text{FPR})$ understates harm: the absolute FPR still rises, so the negative value reflects dispersion reduction, not reduced alarm burden. Absolute clean $\rightarrow$ poisoned FPR at $f=0.4$ (Table 3 clean column): Global $0.041 \rightarrow 0.043$, Local $0.043 \rightarrow 0.046$, Cluster $0.043 \rightarrow 0.049$.

Mechanism and conditionality. Lowering a $q=0.95$ quantile requires displacing scores in the upper-tail support; random position replacement achieves this with probability $\approx f$ (40% at $f=0.4$), leaving 60% of upper-tail support intact. The Global policy compounds this via $1/N$ dilution. These results are presented as a lower bound under Replace-Fixed-Budget; targeted upper-tail removal would achieve larger shifts.

5.4 Policy-Differentiation Summary

Table 5. Global propagates diluted threshold shifts federation-wide, while Local confines them to the victim ($f=0.4$, $n=10$ seeds). Blast counts Gate-1-material non-victim threshold shifts, not measured downstream harm. [†] Shared threshold; [‡] k -means churn, mean non-victim $\Delta\tau=0.0003$, std = 0.017.

Policy	Victim $\Delta\tau$	Victim ΔTPR (pp)	Non-vic. $\Delta\tau$	Blast
Global	+0.100	−6.1	+0.100 [†]	$\approx 8/8$
Cluster	+0.899	−4.9	$\approx 0.000^{\ddagger}$	$\approx 6/8$
Local	+0.902	−7.9	0.000	0/8

Table 5 quantifies the threshold-propagation contrast under the raising attack at $f=0.4$. Global propagates the threshold shift to all nine clients: the non-victim $\Delta\tau$ equals the victim $\Delta\tau$ by construction (shared threshold), and the blast count is 8/8 non-victims in nearly every seed. Cluster produces near-Local victim-level threshold shifts (+0.899 vs. +0.902), though victim ΔTPR remains smaller (−4.9 vs. −7.9 pp); non-victim threshold changes are small in magnitude (mean $\Delta\tau \approx 0.0003$, std = 0.017, driven by k -means cluster reassignment rather than aggregation), with 6/8 non-victims showing Gate-1-material but small-magnitude threshold perturbations. The blast count measures threshold materiality, not downstream harm: non-victim rates were not recomputed for Cluster. Its mean non-victim $\Delta\tau$ is two to three orders of magnitude below the weakest victim shifts ($f=0.1$: +0.033 Global, +0.253 Cluster, +0.298 Local; Table 4), so large non-victim ΔTPR changes are implausible but unverified. Global, by contrast, transmits $\Delta\tau$ to every non-victim, though its downstream effect is

likewise inferred rather than measured. Local yields the highest per-victim shift with zero non-victim blast by policy design.

6 Discussion and Limitations

Score-level proxy and detectability. The experiment operates at the score level: reconstruction scores in the calibration buffer are replaced programmatically rather than through generated traffic. A complete end-to-end attack would additionally require generating network packets whose autoencoder reconstruction error lands in the target tail of the victim’s score distribution, which is hard: it requires controlling feature-level statistics through the encoder-decoder nonlinearity under real network behavior. This work is a *vulnerability-surface characterization*, not a deployed attack; traffic-level realizability is future work. A related implementation limitation: with-replacement sampling from the tail reservoir at $f=0.4$ produces duplicate scores in the calibration buffer—for the smallest per-client calibration set ($n_v \approx 2,622$), on the order of 10^3 expected collision pairs, easily flagged by a uniqueness check; evading this requires score perturbation at the cost of leaving the exact tail boundary. Detectability remains an open problem.

Global propagation and lowering bounds. The $1/N$ -diluted Global shift is definitional; what is empirical is that the residual magnitude remains measurable above the seed-to-seed variance. The Local/Cluster lowering results are a *lower bound* under Replace-Fixed-Budget (§5): adversaries with access to the buffer’s rank structure could achieve larger threshold displacements and alarm-burden effects. These results reveal a trade-off in policy design: Local maximizes per-client isolation at the cost of the largest per-victim exposure, whereas Global limits per-victim shift via averaging but propagates residual threshold shifts federation-wide; defense designers must choose which risk to bound.

Scope limitations. Claims are bounded to N-BaIoT (9 clients), the autoencoder architecture, FedAvg, and $q=0.95$; generalization to other datasets, detectors, FL algorithms, larger federations, and quantiles is open. The experiment assumes a single compromised client; multi-client collusion is a direct extension. The $n=10$ seed bootstrap CI coverage may differ from nominal at the tails. Defense design for the calibration channel (threshold validation, calibration-set integrity constraints) is out of scope and merits independent study.

7 Conclusion

This paper characterized the post-training threshold-calibration stage in federated IoT anomaly detection as an attack surface that the studied training-phase and aggregation-phase defenses do not address. Under a score-level contamination model on the N-BaIoT nine-device federation (FedAvg autoencoder, $q=0.95$, single compromised client), the raising attack degrades victim detection rates by 2.4–7.9 pp across non-zero injection fractions (4.9–7.9 pp at $f=0.4$)

with 100% Gate-1 pass rates and 95% bootstrap CIs excluding zero. The lowering attack shifts thresholds downward, raising ΔTPR by 10–12 pp and the primary alarm-burden metric $\Delta\text{CV}(\text{FPR})$ by 0.05–0.06 for Local and Cluster (at $f=0.4$, a lower bound under Replace-Fixed-Budget). The ΔTPR increase reflects operating-point displacement rather than a security benefit. The policy ordering—per-victim exposure maximized under Local, diluted but federation-wide threshold propagation under Global, small intra-cluster threshold perturbations under Cluster—implies calibration-stage hardening should be matched to the threshold policy, not only the training-phase defense. These findings isolate a calibration-channel vulnerability mechanism, not a complete traffic-generation attack. Future work should address traffic-level realizability, calibration-stage integrity defenses, and generalization across detector architectures and datasets.

Reproducibility. The released artifact bundle is archived on Zenodo under DOI 10.5281/zenodo.21196568. It contains the experiment configuration, paired training/poisoning/analysis seeds, the split manifest, per-cell results, negative-control records, isolation checks, and figure/table scripts. Model checkpoints, hyperparameters, and analysis gates were fixed before poisoned runs; no poisoned metric was used for checkpoint, hyperparameter, or gate selection.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

References

1. Arthur, D., Vassilvitskii, S.: k-means++: The advantages of careful seeding. In: SODA. pp. 1027–1035 (2007), <https://dl.acm.org/doi/10.5555/1283383.1283494>
2. Bagdasaryan, E., Veit, A., Hua, Y., Estrin, D., Shmatikov, V.: How to back-door federated learning. In: AISTATS. pp. 2938–2948. PMLR (2020), <https://proceedings.mlr.press/v108/bagdasaryan20a.html>
3. Belenguer, A., Pascual, J.A., Navaridas, J.: A review of federated learning applications in intrusion detection systems. *Computer Networks* **258**, 111023 (2025). doi:10.1016/j.comnet.2024.111023
4. Bhagoji, A.N., Chakraborty, S., Mittal, P., Calo, S.: Analyzing federated learning through an adversarial lens. In: ICML. pp. 634–643. PMLR (2019), <https://proceedings.mlr.press/v97/bhagoji19a.html>
5. Biggio, B., Nelson, B., Laskov, P.: Poisoning attacks against support vector machines. In: ICML. pp. 1467–1474 (2012), <https://icml.cc/2012/papers/880.pdf>
6. Blanchard, P., El Mhamdi, E.M., Guerraoui, R., Stainer, J.: Machine learning with adversaries: Byzantine tolerant gradient descent. In: NeurIPS. vol. 30, pp. 119–129 (2017), <https://papers.nips.cc/paper/6617-machine-learning-with-adversaries-byzantine-tolerant-gradient-descent>
7. Cao, X., Fang, M., Liu, J., Gong, N.Z.: FLTrust: Byzantine-robust federated learning via trust bootstrapping. In: NDSS (2021). doi:10.14722/ndss.2021.24434
8. Ghosh, A., Chung, J., Yin, D., Ramchandran, K.: An efficient framework for clustered federated learning. In: NeurIPS. vol. 33, pp. 19586–19597 (2020), <https://proceedings.neurips.cc/paper/2020/hash/e32cc80bf07915058ce90722ee17bb71-Abstract.html>

9. Khang, T.T., Duc, T.H., Huynh, D.V., Pham, V.H., Duy, P.T.: P4P: A probe-guided anti-poisoning defense for federated learning-based intrusion detection in IoT networks under non-IID data. *Journal of Network and Computer Applications* **251**, 104502 (2026). doi:10.1016/j.jnca.2026.104502
10. Kloft, M., Laskov, P.: Security analysis of online centroid anomaly detection. *Journal of Machine Learning Research* **13**(118), 3681–3724 (2012), <https://www.jmlr.org/papers/v13/kloft12b.html>
11. Laridi, S., Palmer, G., Tam, K.M.M.: Enhanced federated anomaly detection through autoencoders using summary statistics-based thresholding. *Scientific Reports* **14**(1), 26704 (2024). doi:10.1038/s41598-024-76961-2
12. McMahan, B., Moore, E., Ramage, D., Hampson, S., Agüera y Arcas, B.: Communication-efficient learning of deep networks from decentralized data. In: *AISTATS*. pp. 1273–1282. PMLR (2017), <https://proceedings.mlr.press/v54/mcmahan17a.html>
13. Meidan, Y., Bohadana, M., Mathov, Y., Mirsky, Y., Shabtai, A., Breitenbacher, D., Elovici, Y.: N-BaIoT—network-based detection of IoT botnet attacks using deep autoencoders. *IEEE Pervasive Computing* **17**(3), 12–22 (2018). doi:10.1109/MPRV.2018.03367731
14. Quyen, N.H., Duy, P.T., Nguyen, N.T., Khoa, N.H., Pham, V.H.: FedKD-IDS: A robust intrusion detection system using knowledge distillation-based semi-supervised federated learning and anti-poisoning attack mechanism. *Information Fusion* **117**, 102807 (2025). doi:10.1016/j.inffus.2024.102807
15. Rey, V., Sánchez Sánchez, P.M., Huertas Celdrán, A., Bovet, G.: Federated learning for malware detection in IoT devices. *Computer Networks* **204**, 108693 (2022). doi:10.1016/j.comnet.2021.108693
16. Sáez-de-Cámara, X., Flores, J.L., Arellano, C., Urbietta, A., Zurutuza, U.: Clustered federated learning architecture for network anomaly detection in large scale heterogeneous IoT networks. *Computers & Security* **131**, 103299 (2023). doi:10.1016/j.cose.2023.103299
17. Sattler, F., Müller, K.R., Samek, W.: Clustered federated learning: Model-agnostic distributed multitask optimization under privacy constraints. *IEEE Transactions on Neural Networks and Learning Systems* **32**(8), 3710–3722 (2021). doi:10.1109/TNNLS.2020.3015958
18. Thein, T.T., Shiraishi, Y., Morii, M.: Personalized federated learning-based intrusion detection system: Poisoning attack and defense. *Future Generation Computer Systems* **153**, 182–192 (2024). doi:10.1016/j.future.2023.10.005
19. Tolpegin, V., Truex, S., Gursoy, M.E., Liu, L.: Data poisoning attacks against federated learning systems. In: *ESORICS*. pp. 480–501. Springer (2020). doi:10.1007/978-3-030-58951-6_24
20. Yin, D., Chen, Y., Ramchandran, K., Bartlett, P.: Byzantine-robust distributed learning: Towards optimal statistical rates. In: *ICML*. pp. 5650–5659. PMLR (2018), <https://proceedings.mlr.press/v80/yin18a.html>